AI 321: Machine Learning for Sequential Data

Linear Dynamic Systems

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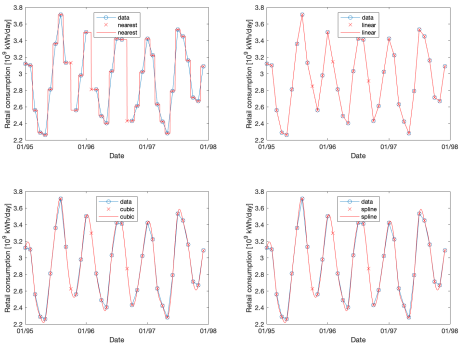
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Overview

1. Waveform fitting vs system identification
2. Why study linear systems?
3. A realistic example
4. What are the different system modes?
5. Input Design

Example of waveform fitting

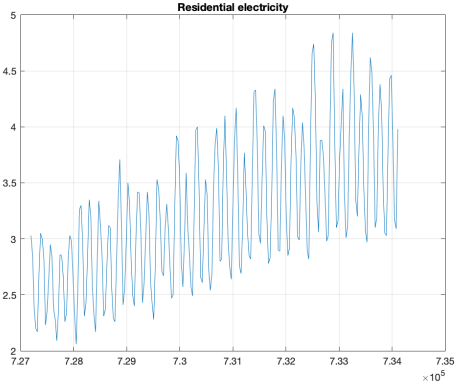
- Real world data have some missing values, some interpolation



- Data cleaning needs to be done. Some smoothing filter, i.e convolution or FIR filter.

Modeling the data by some function

- After interpolation and smoothing.



- We want to fit a model to the data, i.e. polynomial function + sinusoid component.

Model fitting

- Compute the FFT to estimate ω, i.e. $\omega = \frac{2\pi T}{365}$
- Start with an initial linear model + sinusoid component

$$y = C_0 + C_1 t + C_2 \cos(\tfrac{2\pi T}{365} t)$$

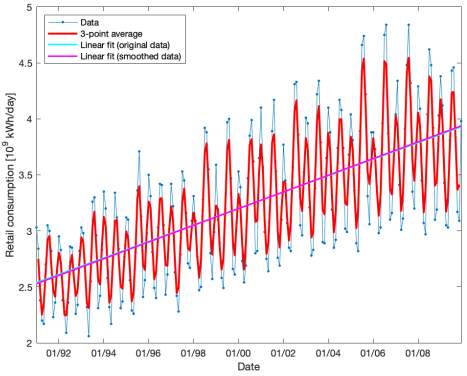
$$\begin{bmatrix} 4.03 \\ 3.84 \\ 3.38 \\ \vdots \\ 3.92 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0.001 & 0.4831 \\ 1 & 0.002 & -0.4431 \\ 1 & 0.003 & -0.9990 \\ \vdots & \vdots & \vdots \\ 1 & 1 & 0.4907 \end{bmatrix} \begin{bmatrix} C_0 \\ C_1 \\ C_2 \end{bmatrix}$$

- From the data, solve the linear regression problem.

$$y = Fc; \quad c = F \backslash y$$

Model fitting

- Linear polynomial fit

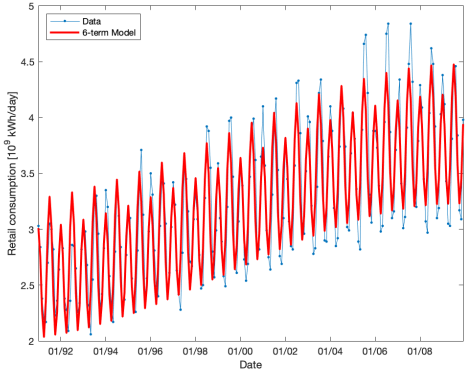


- Can improve the model by adding more polynomial terms

Model fitting

- 6-parameter model

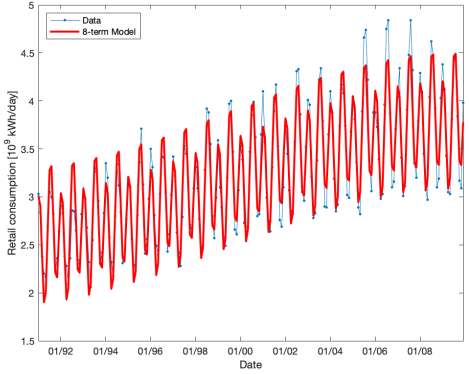
$$y = C_0 + C_1 t + C_2 t^2 + C_3 t^3 + C_4 \cos(\tfrac{2\pi}{365} t) + C_5 \cos(\tfrac{4\pi}{365} t)$$



Model fitting

- 8-parameter model

$$y = C_0 + C_1 t + C_2 t^2 + C_3 t^3 + C_4 \cos(\tfrac{2\pi T}{365} t) + C_5 \cos(\tfrac{4\pi T}{365} t) + C_6 \sin(\tfrac{2\pi T}{365} t) + C_7 \sin(\tfrac{4\pi T}{365} t)$$



- Waveform fitting is different from system identification (data predicts the model).

Why Study Linear Systems?

Applications arise in many areas

- automatic control systems
- communications and signal processing
- economics and finance
- circuit analysis, simulation and design
- mechanical systems
- aeronautics
- automobile design and simulation
- navigation and guidance

Application are limited by the available computing power.

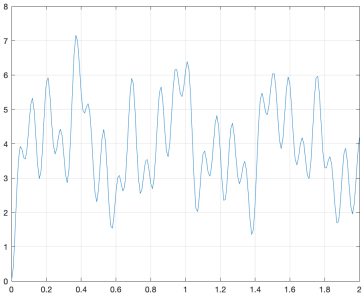
Consider the scenario

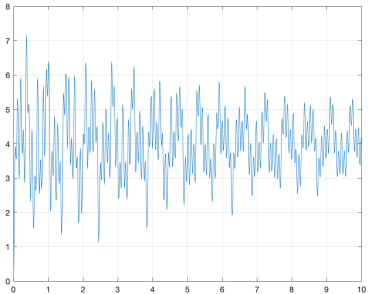
- Let us consider an 8-state single output autonomous system (no input).

$$\dot{x} = Ax, \qquad y = Cx$$

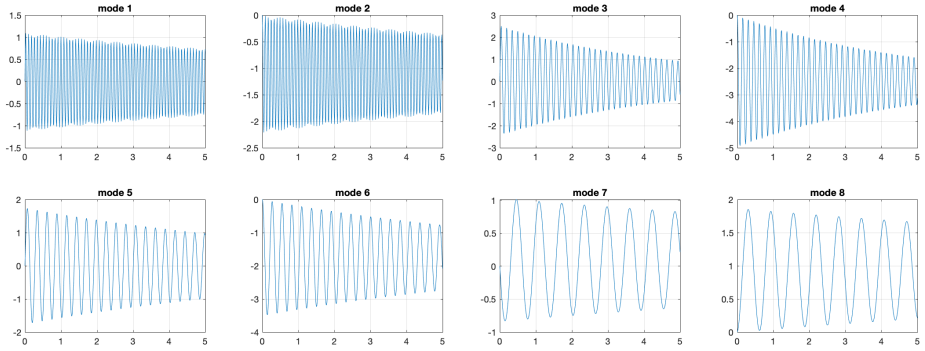
with $x(t) \in \mathbb{R}^8$ and $y(t) \in \mathbb{R}$

- typical output (step response) at different time-scales.





Output decomposition



- Simpler to analyze

Steady State Response to a Step Input

- Input design. Consider a system with input.

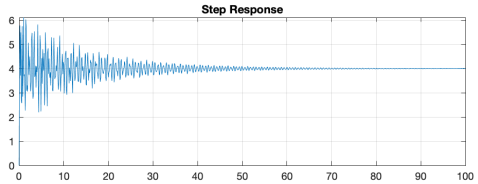
$$\dot{x} = Ax + Bu, \qquad y = Cx$$

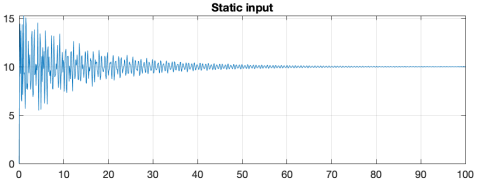
with $x(t) \in \mathbb{R}^8$, $y(t) \in \mathbb{R}$ and $u(t) \in \mathbb{R}$

Determine the appropriate input $u : \mathbb{R}_+ \to \mathbb{R}$ such that $y(t) \to y_d = 10$. Assume $x(0) = 0_{8x1}$.

- This can be solved by

$$u_{static} = (-CA^{-1}B)^{-1} y_d = 2.50$$

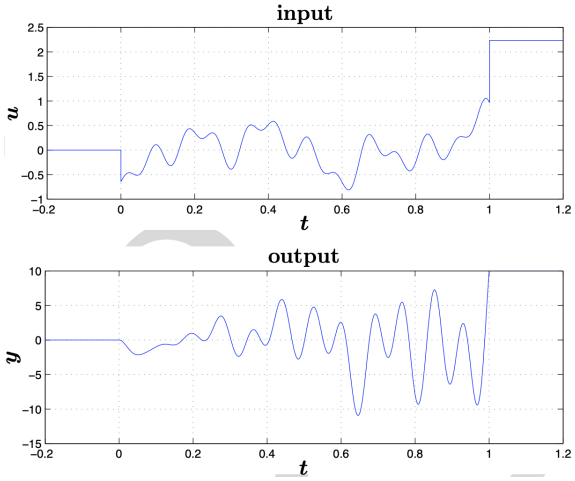




- The output takes about 60s to reach steady-state. Can we do better?

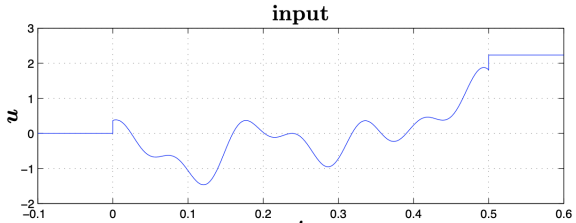
Input Design

- We can do better by applying the following input signal.
- The output converges to y_d in exactly 1s.
- Can we do better?



Input Design

- We can tweak the input signal, and apply a larger input
- The output converges to y_d in exactly 0.5s.
- Larger inputs will make the system converge faster.



System Identification vs Waveform fitting

- System Identification: builds a model of the entire system's dynamic behavior from data, how inputs influence the model output.
  - Natural ability to model multiple inputs and outputs
  - Robustness of the analysis and prediction to different inputs
  - Able to model noise and other disturbances to the system
  - Manages the causal relationship of the inputs to the output.
- Curve fitting: optimizes the parameters of a model to a specific signal or waveform over time. Models static relationships of $y = f(x)$
  - In some cases this is sufficient to make predictions, i.e weather prediction, seismic data analysis, etc.